Karthikeya V S Shanmukha Sankar Vemuluri

M.Tech Artificial Intelligence

ACADEMIC PROFILE

Masters in Technology	7.9	Indian Institute of Science	2026
Bacherlors of Technology	7.53	Sardar Vallabhbhai National Institute of Technology	2023

INTERNSHIPS — 1 Months

Samsung Research Institute of Delhi — **AI Research Intern** — **May'25-Jul'25**

- – Fine-tuned audio models (**BEATs, HuBERT, Wav2Vec, YAMNet**) for 35-class classification.
  – Applied **LoRA, feature matching,** and optimized **Softmax/Sigmoid** activations for predictions.
  – **Distilled** BEATs into YAMNet to enhance lightweight audio recognition.

PROJECTS

Multi-Modal Fashion Recommender System

- Built a multimodal recommendation model combining **ViT and BERT** on H&M Dataset
- Introduced **cross-modal attention** to enhance user preference representation via visual–textual fusion.
- Attained **NDCG@10 ≈ 0.9**, demonstrating strong performance in multimodal recommendation tasks.

Agentic Scholar Research RAG

- Developed an Agentic RAG pipeline with independent agents retrieving research papers, scholarly blogs, top web content, and user-uploaded documents, leveraging Qwen 3 0.6B embeddings and Qwen 2.5 7B instruct for a fully free stack.
- Implemented **iterative query reformulation, reranking**, and FAISS-based vector search to ensure high-precision semantic retrieval.
- Designed an autonomous **multi-agent workflow supporting retrieval validation, and coherent answer synthesis.**

Neural Machine Translation using Transformer

- Developed an Eng–Hin translation model using a **Transformer** architecture with **Byte Pair Encoding** (BPE) tokenization from scratch.
- Implemented **Greedy** and **Beam** Search decoding for optimized translation generation.
- Performed attention head importance analysis and **pruning** to enhance model efficiency with minimal accuracy loss of ~2%.

Multilingual Neural Machine Translation for Low-Resource Indic Languages

- Fine-tuned the pretrained mBART model using adapter-based techniques (LoRA and bottleneck layers) to efficiently adapt multilingual translation for low-resource Indian languages.
- Investigated transfer learning performance on languages absent from pretraining data.
- Evaluated translation quality using BLEU scores to quantify improvements.

Unlearning in Diffusion Models

- Developed a scalable framework for erasing unwanted visual concepts from text-to-image diffusion models while preserving overall image quality.
- Utilized LoRA-based fine-tuning and cross-attention refinement to selectively suppress target concepts without retraining the entire model.
- Designed experiments to evaluate erasure effectiveness and utility retention across object, celebrity, and artistic style categories.

SKILLS

- **Programming Languages:** C, C++, Python, SQL
- **MachineLearning &DeepLearning:** PyTorch,scikit-learn, Transformers, Retrieval-Augmented Generation (RAG), LangChain
- **NLP &AI Tools:** Hugging Face, NLTK, Text, Image, and Speech Processing
- **Version Control & Documentation:** Git, GitHub, LaTeX, Jupyter Notebook

COURSES

Linear Algebra, Probability, Optimisation, DSA (PG)
Intro to NLP, MLSP, AIP (PG)
Neural Networks, Machine Learning, Deep Learning, Speech, DIP (UG)

CERTIFICATIONS

Data Science, Machine Learning and AI using Python	IIT Ropar	2021
PythonProgramming	TwowaitsTechnologies Pvt. Ltd.	2021

ACHIEVEMENTS

- Secured AIR 27 in GATE Data Science and AI (2024).
- Secured AIR 236 in GATE Computer Science (2024).
- Winner of GRP 2K19.

POSITION OF RESPONSIBILITY

- Member, Drishti Club (2019–2021), contributing to Line Follower and RC Robots.
- Mentor, Drishti Club (2022), guiding peers on various projects.

LANGUAGES

English, Telugu, Hindi